Shot Make Probability Model

OKC Thunder - Analyst Intern Project 2026

Serigne N'Dione

An issue with treating expected shot percentages as a solved problem in modern analytics is that everyone usually knows we're using static coordinates, a single closest defender distance metric, and a simplistic tracking-data closeout speed...and they don't just know it, they know you're using it to pretend a 14-foot floater from a leaning guard who just absorbed a shoulder-check at the free-throw line is identical to a 14 foot floater from a wide-open guard walking into a set-shot out of a calm secondary break.

This is the difficult part of spatial modeling: the very implication of a standardized shot-quality metric is that the hardwood is a uniform canvas, when in reality it's this moving, suffocating grid of dynamic physics where a 0.4 second delay in secondary rotation turns a pristine 68% look into a 38% attempt, and the player taking it is always 31, which in the context of rim-penetration is less an age than a terminal diagnosis. Because you can be 31 and 37 at the same time…ask anyone tracking the rim-frequency charts of athletic guards whose first-step speed has dropped by four-tenths of a mile per hour, or any analyst watching a once-dominant finisher start relying on off-pike short mid-range floaters.

This is after looking a smidge slower getting past high-gap digs all season despite the surface-level scoring average matching that All-NBA year, and he's up for a contract extension, and the free-throw rate is down and the restricted-area attempts are down, but the pull-up jumpers are up and the high-risk skip-passes are way up, meaning the volatility is way up, because the player is already grasping for new ways to generate spatial stress, if not through his own hips, then through the desperation of weak-side defenders over-committing to a phantom threat.

Rewind ten or so years and the questions being asked by spatial metrics were almost entirely positional:

Can you make a shot from here?

At a high enough volume to warrant a closeout?

Can you stand here? Can you stand in the weak-side corner? Over there in the slot? Even there on the break?

Can you suppress the urge to take three dribbles into a mid-range contest?

Can you be fine with maintaining spatial width until the lead ball-handler spoon-feeds you an uncontested corner catch?

Can you watch another player dribble for seventeen seconds and remain emotionally available when the pass hits your hands with 2.4 seconds on the shot clock?

All extremely useful questions. Basically, the SAT for modern spatial role players. But it isn't an equal sampling of basketball physics, and like the SAT, the test has so many built-in biases toward static positions and simple end-states rather than the relational friction that produced them.

Consider May of 2016. The Golden State Warriors came off of a 73-win regular season with the highest relative offensive rating of the last twenty years (+8.1), anchored defensively by Draymond Green's hyper-

communicative help-side coverage and a switching infrastructure designed to turn high-level offenses into frantic, low-efficiency perimeter probes.

For 82 games, the Warriors operated with a kind of immaculate defensive serenity: biding their time, baiting ball-handlers into wide-drift arc drives, a preset backline, and closing out with synchronized precision from length-defenders like Klay Thompson, Andre Iguodala…

Then came Russell Westbrook in the Western Conference Finals, a series that remains the ultimate blueprint for how physical acceleration shatters defensive architecture.

Across games against peak Golden State, Westbrook posted raw efficiency numbers that look like an average high-volume guard on a standard stat sheet: 42.0% FG, 36.4% 3PT, and 27.3 points and 11.8 assists per game.

Yet Westbrook was the only force on the planet capable of forcing peak Golden State to consciously move off its preferred chaos-ball. He solved their defensive shell by out-calculating it, and he broke it by out-pacing their historic pace.

A methodical scorer like James Harden attacked its defense a round prior through controlled deceleration, which allowed Golden State's backline to load its weight, pre-align its hips, and stunt into driving gaps while staying attached to weak-side shooters. Westbrook presented a forward-momentum that constantly pulled his body toward aggressive, violent decisions.

He challenged a screen's spatial footprint and motion blur at maximum velocity and forced primary and secondary defenders into immediate, high-stress reaction cycles.

Why did Westbrook's penetration compromise Golden State's defense in ways traditional efficiency models fail to capture? It was his variability of entry velocity and constant line-winning angles.

Harden entered the paint on a controlled arc, which allowed Draymond Green to sit at the edge of the restricted area on a set foot, timing his contest while weak-side defenders maintained their recovery vectors.

Westbrook entered the paint at 14 miles per hour straight down the slot, receiving the ball on a sync cut or penetrating against a defense whose interior rotation was mid-stride. He moved away from shots or passes against a set Golden State shell; and towards attacks against a corrupted shell, a defensive line whose feet were turned, whose hips were open, and whose contact leverage was entirely neutralized before the shot attempt even reached its apex.

This kinetic disruption was so severe that it insulated Oklahoma City's non-spacing lineups. OKC was able to run multi-big lineups featuring Enes Kanter, Steven Adams, Andre Roberson, and Serge Ibaka without having to the traditional half-court spacing tax.

The defense was so over-committed to halting Westbrook's downhill momentum that its ability to contest secondary dump-offs and offensive rebounds collapsed.

And then the 3-1 comeback happened…

But that is the distinction this model formalizes: Shot probability is less of a function of where a player stands when he releases the ball, but a relational ecosystem governed by how aggressively the initial attack forces a defensive shell off its balance lines, how rapidly time strips away secondary recovery options, and whether the offense generates a corrupted, scrambling environment or settles against an intact, pre-set wall.

To move beyond static spatial assumptions and measure the dynamic physics of shell corruption, this paper constructs a multi-tiered feature pipeline fed directly into an XGBoost Gradient Boosting framework. The model captures non-linear interactions across the full kinetic lifecycle of a possession, synthesizing drive geometry, option decay, lane congestion to shooter & defender zone profiles and tendencies before finally moving to shot quality and true shot-making efficiency beyond expected indices.

The model processes raw spatial and temporal tracking arrays through seven layers of a possession.

1. Option Decay Index

A defense begins with a simple, testing question: what happens if we push here? When the offense answers, the defense takes that answer, files it away, and prepares a counter for the next time the offense tries the exact same action. Over thousands of possessions, a elite defense stops thinking in rigid rules. It stops panicking when a rotation breaks. It starts reading the floor in shades and textures, reflexively knowing where the ball wants to go before the ball handler even knows himself.

The Option Decay Index ODI measures the rate at which this defensive memory suffocates an offense as the shot clock ticks down. At 18 seconds, a ball-handler has options. At 3.8 seconds, not so much. He probably would rather not take a contested, off-balance 14-foot floater. But he's taking it nonetheless because a defense that has spent 82 games reflexively training itself to compress space on the floor has practically deleted every other escape route on the floor.

Table 1: Make Rate by Option Decay Index — Training Set (n = 425,719)

Bucket	Shot Clock	Make Rate	vg SQ	n
Idealized / Ambiguous	≥18s	53.1%	0.384	72,615
Probing / Conditional	8–17s	45.8%	0.379	258,800
Narrowed / Predictable	4–7s	43.1%	0.368	56,671
Panic / Instinctive	<4s	36.2%	0.332	37,633

Shot Quality drops only 5.2 pts from Idealized to Panic. Make rate drops 16.9 pts. That gap, or lack thereof, is the decision deterioration story…the shot quality doesn't collapse, so it leans towards the the lack of options as the primary driver of inefficiency .

2. Spatial Context: Zone and Relative Zone Distance

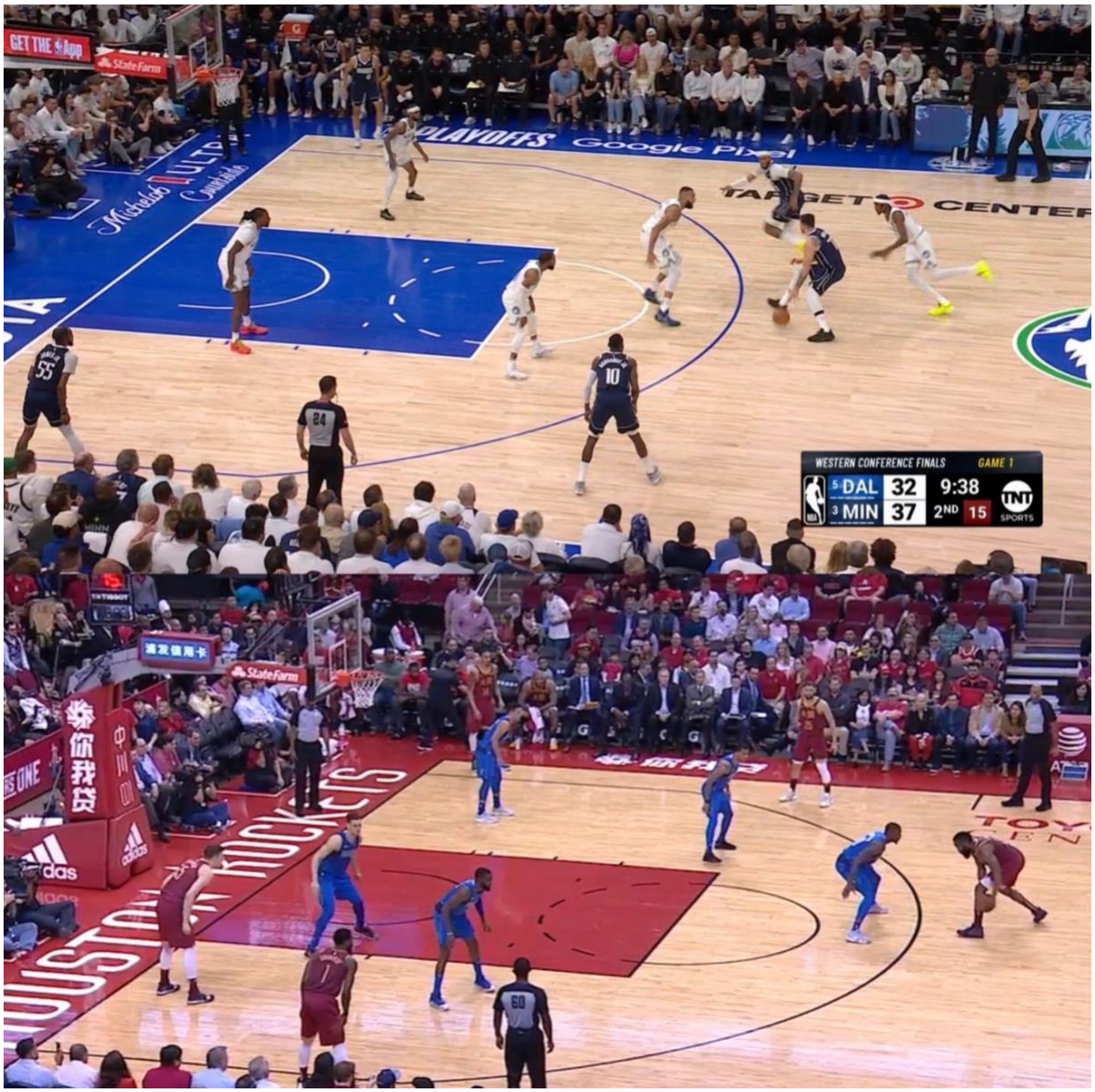
A 6-foot finish at the rim is a high-value *asset*. A 10-foot floater in the short mid-range is a touch-based *compromise*. A 19-foot pull-up in the long mid-range is a analytical *concession*, especially if your head coach is Kenny Atkinson. And a deep shot from three changes the risk-reward ratio of the possession. The Zone Distance Relative Metric smooths this gap by calculating a player's continuous position relative to the boundaries of their specific zone. At the rim: it evaluates how far an attempt drifts off the rim center

before it transitions into a soft short mid-range floater. At the short mid range, it measures proximity to the restricted area seam versus being pushed back toward the dead-zone edge. At the long mid-range it separates a comfortable 14-foot rhythm pull-up from an inefficient 21-foot "long-2" taken an inch inside the stripe. Finally, beyond the three point line it captures the steep efficiency drop-off between a clean line-up corner three and a deep 34-foot bomb.

3. Drive Geometry and Lateral Displacement

A line winner maintains a low center of gravity and gets his shoulder directly past the defender's inside hip. When the point-of-attack defender attempts a chest bump, a line winner can absorb the contact through the torso without footwork becoming choppy or the drive bending off course. Because he can win the line cleanly in naked isolation, he declines the tradeoff of a screen: he doesn't need a big man to alter the leverage, which keeps a second rim-protector out of the central paint lane, ultimately freeing up lane congestion and higher efficiency shots

When a ballhandler cannot consistently hold that linear plane through contact, the initial bump turns their chest outward toward help. Over time, losing that rim-pressure ability conditions players away from Tony Parker-style efficiency of movement and forces them to rely on decelerations, step-backs, floaters, and heavy pick-and-roll screening to manufacture artificial leverage. Under normal circumstances, getting bump-deflected off a direct linear drive is a defensive win, a player is forced into a lower-efficiency lateral escape. However there exists a rare class of elite mid-range creators, the Luka Doncic's and Shai Gilgeous-Alexander's, who turn this standard defensive victory on its head and live in these sharp decelerations and side steps along that wider angle and create massive pockets of space.



Luka Doncic & James Harden— Line Winning vs Screen Tradeoff

Table 3: Make Rate by Drive Geometry — Training Set (n = 425,719)

Geometry	Lateral Displacement	Make Rate	Avg SQ	Gap	n
Line Winner	≤2.0 ft	63.9%	0.321	–0.318	77,717
Slight Drift	2.1–5.0 ft	55.3%	0.303	–0.250	77,226
Wide Driver	>5.0 ft	38.4%	0.245	–0.139	23,660
Perimeter Release	≥10 ft from rim	37.9%	0.426	+0.048	247,116

Line Winner outperforms its own shot quality by 31.8 pts while Wide Drift has the worst shot quality out of the 4.

4. Lane Congestion and Shell Penetration

Lane Congestion is the tax a drive pays when entering the paint, quantified by the real-time density of defensive players within a driver's immediate path. When this metric spikes before a handler reaches the paint, the floor compresses and drive efficiency collapses into forced pickups and off-balance floaters. Highly congested lanes are often self-inflicted by unnecessary pick-and-rolls, which invite a second heavy rim protector into the central channel. Conversely, elite line-winners bypass screens to preserve clean corridors, while offenses that space dynamic cutters out-leverage static help, keeping lane density low and allowing drivers to navigate open air rather than wading through a pre-rotated shell.

Table 4: Make Rate by Lane Congestion — Training Set (n = 425,719)

Level	Defenders in Lane	Make Rate	n
Clear Lane	0	62.1%	46,007
Single Coverage	1	44.1%	310,009
Shell Forming	2	43.1%	65,637
Shell Locked	3+	36.9%	4,066

Shell Penetration measures whether a drive cracks the foundation of a defense or just grazes its outer boundary. This is why the precise pickup point…where a dig or stunt occurs…is often more critical than the overall scheme itself.

On a wide drive against a defense playing high in the gaps, a dig arriving just below the three-point line asks a ballhandler to solve a radically different puzzle than a contest at the nail:

When a driver reaches eleven feet, the offense has already migrated deep into the defensive shell. The low-man has committed, the primary rim protector has declared his rotation, and secondary defenders have already drifted into no-man's land with their feet turned. Even if the driver gathers early, the shell is already broken—somebody is out of position, and the worst-case outcome is a dump-off pass or short-midrange pull-up or that creators are increasingly getting more and more comfortable with

Picking up further up the court diminishes the drive before any pressure reaches the backline. Against extra-large, 6-foot-8 point guards who see over everything, high-gap positioning makes help visible early, offering a "good but sterile" answer: take a contested pull-up from twenty feet or kick it out to a non-Klay Thompson role player outside the arc. The defense isn't trying to make the star choose incorrectly but rather it's forcing him to make the correct read so early that his downstream role players are forced to solve the remaining 14 seconds of possession logic themselves.

Russ's genius was refusing to take these sterile, early answers. By leveraging playmaking over scoring: defenses had to account for every passing window, with each partial-advantage having 10 passing options attached instead of 1-2 scoring ones, to the point where the possession itself became a load-bearing advantage.

Table 5: Make Rate by Shell Corruption — Training Set (n = 425,719)

Category	Entry Condition	Make Rate	n
Sync Cut Finish	0 dribbles, <8 ft	68.4%	57,073
Shallow Probe	1–2 dribbles, <6 ft	59.1%	39,265
Deep Corruption	≥3 dribbles, <10 ft	49.1%	68,027
Forced Sterile	≤1 dribble, ≥16 ft	39.6%	4,054
Mid Creation	Other creation	38.1%	117,850
Catch and Release	0 dribbles, ≥8 ft	37.9%	139,450

Sync Cut Finish carries the highest make rate (68.4%) since catch, gather, and shot happen in a split second before the defense turns to contest. Catch and Release has the highest shot quality (0.455) and the lowest make rate (37.9%).

5. Shooter Zone Profiles and Shot Share Archetypes

Consider a guard drifting toward the right elbow with 3.2 seconds on the shot clock. A standard spatial evalutation assigns a sterile, position-agnostic expected value based on every 14-foot jumper on record. It does not know, or care, that he is statically classified as a central rim dependent player, making this mid-range pull-up an out-of-archetype breakdown rather than a comfortable weapon. Within this model's ecosystem, it reads as a high-tax concession under pressure.

To capture this context, our XGBoost model incorporates static player-level taxonomies directly into its feature space via Leave-One-Out (LOO) efficiency deltas across five floor regions: Rim, Short Mid-Range (SMR), Long Mid-Range (LMR), Three-Point Arc, and Tip/Putback.

Within XGBoost, these fixed profiles form non-linear decision splits. The tree structure learns that a 3.8-foot contest against a contest resistant, everywhere-scorer (i.e. the "I'm personally manufacturing most, if not the entire, offense in the halfcourt through my self-creation" job) carries a drastically different expected conversion rate than an identical contest against a conditional system scorer operating outside his baseline profile.

6. Defender Identity & Zone Environment

An anchor big dragged out to twenty-three feet is in a hostile defensive environment. The shot quality numbers makes the real damage plain: step that anchor big out to the three-point arc and he surrenders a staggering 0.514 Shot Quality, the single worst defensive environment in the entire dataset. Place him back at the rim, in his native habitat, and that allowed Shot Quality plunges to 0.366.

By classifying defenders into core identities-anchor big, mobile big, perimeter stopper, and balanced rotator, and mapping them against the shot zone, we measure this hidden leverage. When fed into XGBoost, the tree learns to see past simple defender proximity and adjust expected efficiency the moment a defender is forced to operate outside his native role.

Defender Role	Make Rate	Avg SQ Allowed	n
Anchor Big	46.8%	0.366	91,745
Mobile Big	46.7%	0.375	54,202
Balanced Rotator	45.5%	0.375	211,393
Perimeter Stopper	44.7%	0.384	68,379

Table 6a: Make Rate by Defender Role — Training Set (n = 425,719)

Defender Zone Fit	Make Rate	n
Optimal	44.5%	118,650
Neutral	45.0%	246,218
Hostile	48.0%	60,851

Shots against a defender in a Hostile zone environment convert at 48.0% despite the opponent's best efforts

7. Shot Quality and Shot Making Efficiency

Shot Quality measures the exact structural resistance a possession generates before individual skill enters the frame. It quantifies environmental friction by integrating three spatial variables:

Space (S): Evaluates location continuously, rewarding high-yielding rim and perimeter zones while discounting low-efficiency mid-range zones.

Time (T): A temporal decay factor that penalizes late-clock bailouts inside the final four seconds (t <= 4.0s).

Defensive Suppression (D): A multi-frame defensive blend tracking closeout distance, four-frame acceleration, release slope, and weak-side help proximity, normalized against elite training-set friction. Backcourt heaves set this to zero.

Where typical metrics treat shot making in a vacuum, active zone calculates a shooter's Empirical Bayes-smoothed Shot Making Efficiency relative to the baseline. It establishes the league-average conversion rate given that specific shot quality, then measures how far above or below that expected baseline the shooter converts within that exact zone, extracting authentic shotmaking talent.

Model Architecture

The engineered feature pipeline feeds into an XGBoost gradient boosted tree ensemble, chosen specifically for its ability to learn complex, non-linear interactions across high-dimensional tracking features without requiring manual interaction terms.

The final feature matrix partitions 11 continuous variables and 10 categorical variables. Categorical are one-hot encoded, with unseen factor levels mapped to UNKNOWN. Missing values in continuous columns are imputed using training-set medians calculated strictly prior to test data to prevent leakage.

Key Hyperparameters

eta = 0.01: Slow learning rate to avoid overfitting, compensated by 2000 rounds; the model converged at round 1942 on the holdout.

max_depth = 6: Enough depth to capture multi-layer interactions without memorizing noise.

colsample_bytree = 0.4: Blinds each tree to 60% of columns, preventing Shot Quality from monopolizing every split.

subsample = 0.8: Row-level subsampling for ensemble diversity.

The final model trains on the full training set with fixed nrounds = 2000. OOS log-loss was evaluated on a held-out 20% split: 0.6266.

What the Model Can't See

Fatigue and foul trouble: A player on his fourth foul in the third quarter operates differently than the same player in a clean first-quarter possession; no tracking signal captures that.

Defensive scheme context: The model knows defender roles but not the defensive call. A drop coverage and a hedge look identical in the tracking data until the big's feet move.

Offensive play design: A mismatch hunt after three actions of screening versus a bailout iso look the same at the moment of release.

Game score / leverage: Gamestate captures transition/halfcourt but not whether it's a 3-point game with 40 seconds left.

Shooter fatigue within a game: Shot making efficiency is season-level; a guard who just took 8 shots in 4 minutes is treated the same as one who's been resting.

To further refine the model's predictive precision, future iterations will expand beyond static point-in-time features to capture broader temporal and structural context. First, the pipeline will incorporate multi-possession sequence features to model how momentum and cumulative defensive attrition impact individual shot outcomes over preceding possessions. The defensive tracking framework will also transition from snapshot proximity metrics to continuous defender closing speeds, to be able to measure acceleration/deceleration during the contest phase. Additionally, defender context will evolve from individual matchups to a lineup-level defensive identity that measures the overall integrity of the five-player unit. Finally, the Empirical Bayes Shot-Making Efficiency (SME) signal will be recalibrated quarterly rather than evaluated annually, allowing the model to dynamically detect mid-season finishing degradation curves in real time.